

Lab Report: Week 2

Determination of the Spring Constant

Chhaya Ramnani (UID: u8336188)

Lab Partner: Candice Ng

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1 Introduction

aim

2 Theory

2.1 Static Method

```
import numpy as np
# Example calculation of spring constant
m = np.array([0.05, 0.10, 0.15]) # mass in kg
x = np.array([0.012, 0.024, 0.037]) # extension in m
k = (m * 9.796) / x
print("Calculated k:", k)
```

2.2 Dynamic Method

3 Data and Results

3.1 Static Data

Mass m (g)	Force mg (N)	Extension x (m)	k (N/m)

3.2 Dynamic Data

Mass m (g)	Mean Period T (s)	T^2 (s ²)

4 Observation and Conclusion